

STAT 801A Lab 2: Quarto and Probability Distributions

Answer Key

2026-09-14

Quarto

An R script (.R) holds only code. A Quarto document (.qmd) holds code, output, and writing in one file. You “render” the file and Quarto runs every piece of R code, captures the results, and stitches everything into a finished HTML page, PDF, or Word document.

To learn more about Quarto see <https://quarto.org>.

Text Formatting

Level 1 heading

Level 2 heading

Level 3 heading (and so on)

italic and ****bold**** and ‘code style’

- bullet point

1. numbered item

2. another item

Make sure to leave a blank line between paragraphs. Two paragraphs jammed together with no blank line will render as one.

Code chunks

We can run normal R code with code chunks. You can insert one with **Ctrl + Alt + I** (**command + option + I** on macOS) or you can insert one manually

```
2 + 2
```

```
[1] 4
```

Chunk options

Options go inside the chunk on lines starting with **#|**. The space after **#|** is required.

```
mean(c(82, 91, 76, 88, 95))
```

```
[1] 86.4
```

Inline code

To drop a computed number into a sentence, use code wrapped in between backticks.

```
scores <- c(82, 91, 76, 88, 95)
```

The mean score was 86.4

Rendering

Click the **Render** button at the top of the editor (or press **Ctrl + Shift + K**, or **Command + Shift + K** on macOS). RStudio saves the file, runs all the code from top to bottom in a fresh R session, and opens the result in whatever format is specified

Math Equations with LaTeX

Quarto renders mathematical notation using LaTeX syntax.

Symbol reference

You want	You type	Result
Greek letters	<code>\mu</code> <code>\sigma</code> <code>\lambda</code> <code>\alpha</code> <code>\beta</code>	μ σ λ α β
Capital Greek	<code>\Sigma</code> <code>\Delta</code> <code>\Phi</code>	Σ Δ Φ
Subscript	<code>x_1</code> and <code>x_{10}</code>	x_1 and x_{10}
Superscript	<code>x^2</code> and <code>e^{-\lambda}</code>	x^2 and $e^{-\lambda}$
Fraction	<code>\frac{a}{b}</code>	$\frac{a}{b}$
Square root	<code>\sqrt{x}</code> and <code>\sqrt[3]{x}</code>	$\sqrt{x}$ and $\sqrt[3]{x}$
Summation	<code>\sum_{i=1}^n x_i</code>	$\sum_{i=1}^n x_i$
Product	<code>\prod_{i=1}^n x_i</code>	$\prod_{i=1}^n x_i$
Integral	<code>\int_{-\infty}^{\infty} f(x) dx</code>	$\int_{-\infty}^{\infty} f(x) dx$
Binomial coefficient	<code>\binom{n}{k}</code>	$\binom{n}{k}$
Bar / hat	<code>\bar{x}</code> and <code>\hat{p}</code>	$\bar{x}$ and $\hat{p}$
Inequalities	<code>\leq</code> <code>\geq</code> <code>\neq</code> <code>\approx</code>	$\leq$ $\geq$ $\neq$ $\approx$
Plus/minus	<code>\pm</code>	$\pm$
Distributed as	<code>X \sim N(\mu, \sigma^2)</code>	$X \sim N(\mu, \sigma^2)$

Wrap math in single dollar signs to put it in a sentence, or double dollar signs to put it on its own centered line.

ggplot2

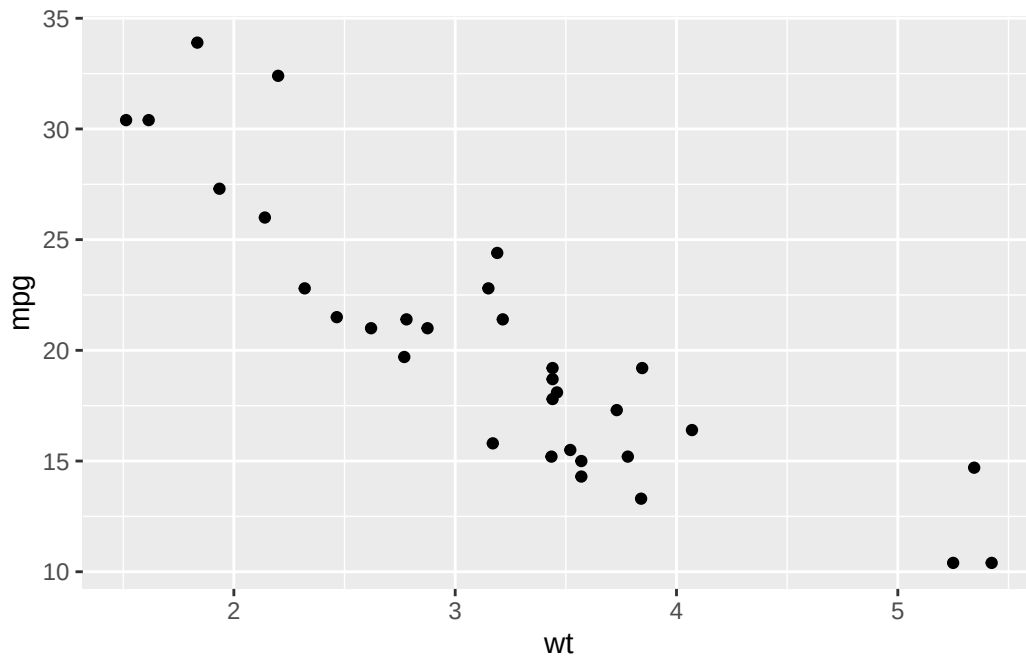
```
#install.packages("ggplot2")

library(ggplot2)
```

Every ggplot needs three things, and layers are joined with a `+`:

1. Data - a data frame.
2. Aesthetic mapping - which column goes on which axis, written inside `aes()`.
3. A geom - what type of plot is drawn

```
ggplot(data = mtcars, aes(x = wt, y = mpg)) + geom_point()
```



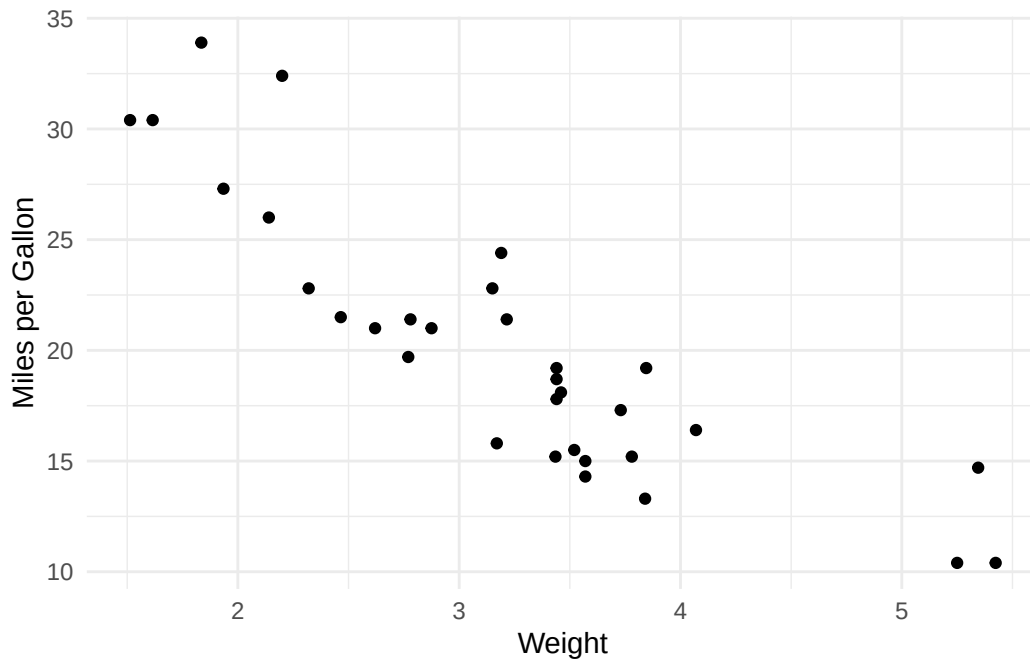
Common geoms

Geom	Draws	Typically needs
<code>geom_point()</code>	Scatterplot	x and y
<code>geom_line()</code>	Connected line	x and y
<code>geom_col()</code>	Bars at heights you supply	x and y
<code>geom_histogram()</code>	Histogram	x only
<code>geom_boxplot()</code>	Boxplot	x and y
<code>geom_smooth()</code>	Trend line	x and y

Labels and themes

`labs()` sets the title and axis labels. A `theme_*()` function changes the overall look.

```
ggplot(data = mtcars, aes(x = wt, y = mpg)) +
  geom_point() +
  labs(x = "Weight", y = "Miles per Gallon") +
  theme_minimal()
```



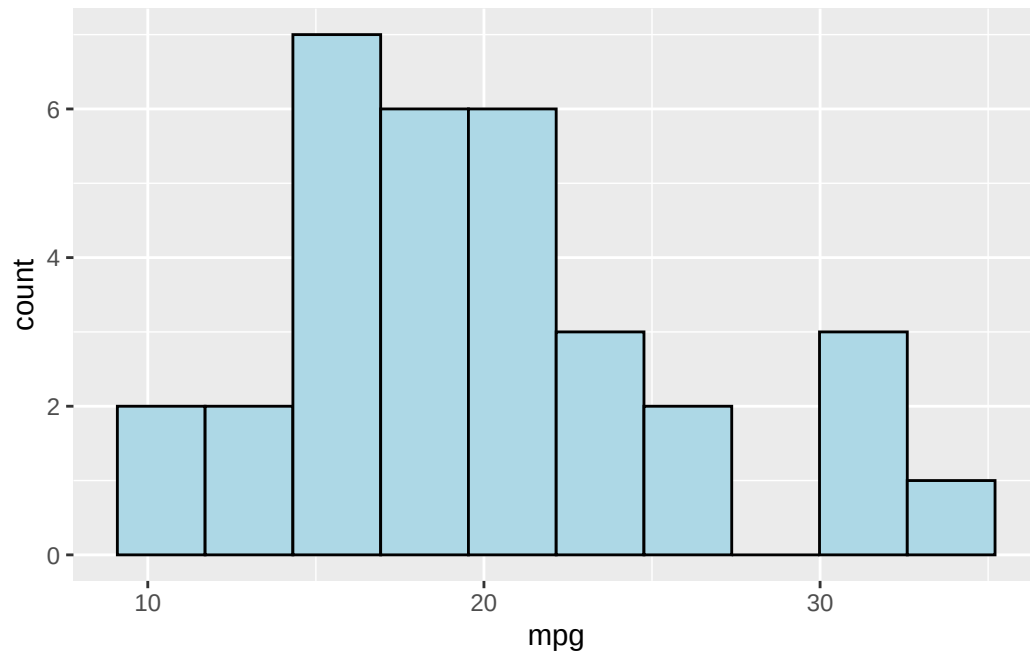
Mapping color vs. setting color

Inside `aes()`, a color is mapped to a variable, so it changes from point to point and a legend appears

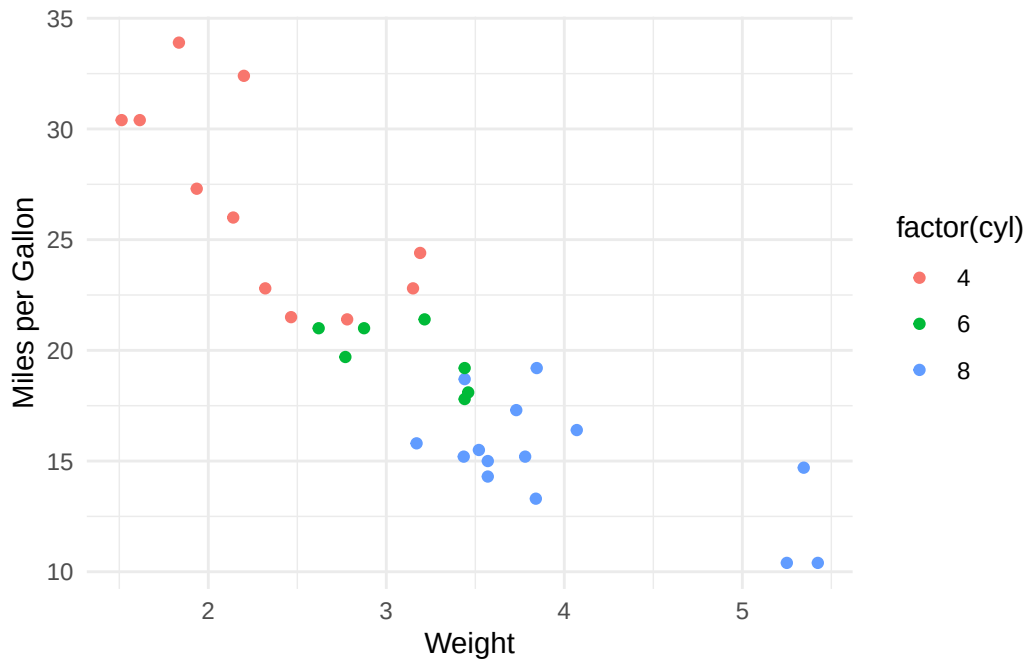
Outside `aes()`, a color is set to one fixed value for everything

Generally, if the appearance should depend on the data, it goes inside `aes()`. If it is just a style choice you are making, it goes outside.

```
ggplot(data = mtcars, aes(x = mpg)) +  
  geom_histogram(bins=10,color = "black",fill="lightblue")
```



```
ggplot(data = mtcars, aes(x = wt, y = mpg, color = factor(cyl))) +  
  geom_point() +  
  labs(x = "Weight", y = "Miles per Gallon") +  
  theme_minimal()
```



Probability Distributions in R

A **probability distribution** is a rule that tells you how likely each possible outcome is.

Discrete:

- Binomial - Number of successes in a fixed number of trials
- Poisson - Count of events occurring in a fixed interval of time or space

Continuous

- Normal - Continuous distribution with a symmetric, bell-shaped curve

When we assume a random variable follows a particular probability distribution, we can calculate the probability that the random variable takes a particular value or falls within a particular range of values.

Wording	Distribution
“out of 20 coin flips,” “12 questions,” “10 free throws”	Binomial
“what proportion of trials succeed”	Binomial
“an average of 4.5 per hour,” “3.2 per month”	Poisson
“how many arrive,” “how many occur”	Poisson

Wording	Distribution
“normally distributed,” “bell-shaped”	Normal
“mean of 500 hours with a standard deviation of 45”	Normal

Notation

You will see distributions written in shorthand. The symbol $\sim$ is read as “is distributed as.” The letter X stands for the thing you are counting or measuring.

Binomial: $X \sim \text{Bin}(n, p)$

“X is distributed binomial with n trials and success probability p .”

Poisson: $X \sim \text{Pois}(\lambda)$

Read: “X is distributed Poisson with rate λ .”

Normal: $X \sim N(\mu, \sigma^2)$

Read: “X is distributed normal with mean μ and variance σ^2 .”

Distribution	Parameters	Meaning
Binomial	n, p	number of trials, probability of success
Poisson	λ	average number of events per interval
Normal	μ, σ	mean, standard deviation

Once we identify the parameters of interest, we can use R to make our lives easier when using these distributions

How R names these functions

R uses a naming system: a prefix describing what you want, plus a root naming the distribution.

The roots:

Distribution	Root
Binomial	binom
Poisson	pois
Normal	norm

The prefixes:

Prefix	What it gives you
d	the probability of exactly one specific value (or density for continuous random variables)
p	the probability of that value or anything smaller
q	works backwards: you give a probability, it returns a value
r	random values drawn from the distribution

Combine the prefix and distribution to get functions:

	Binomial	Poisson	Normal
d	dbinom	dpois	dnorm
p	pbinom	ppois	pnorm
q	qbinom	qpois	qnorm
r	rbinom	rpois	rnorm

The Binomial distribution

A quiz has 12 multiple-choice questions, each with 4 options. A student guesses randomly on every question.

$$X \sim \text{Bin}(12, 0.25)$$

```
n <- 12
p <- 0.25
```

dbinom - exactly one value

Question: What is the probability the student gets exactly 5 correct?

In symbols: $P(X = 5)$

```
dbinom(5, size = n, prob = p)
```

```
[1] 0.1032414
```

There is about a 10.3% chance of getting exactly 5 correct.

Note: R uses “size” and “prob” arguments rather than “n” and “p”

You can ask for several values at once. Giving 0:12 returns the probability of every possible outcome. This gives us the probability distribution

```
dbinom(0:12, size = n, prob = p)
```

```
[1] 3.167635e-02 1.267054e-01 2.322932e-01 2.581036e-01 1.935777e-01  
[6] 1.032414e-01 4.014945e-02 1.147127e-02 2.389848e-03 3.540516e-04  
[11] 3.540516e-05 2.145767e-06 5.960464e-08
```

pbinom - a value or anything smaller

Question: What is the probability the student gets 3 or fewer correct?

In symbols: $P(X \leq 3)$

```
pbinom(3, size = n, prob = p)
```

```
[1] 0.6487786
```

This will yield the same result:

```
sum(dbinom(0:3, size = n, prob = p))
```

```
[1] 0.6487786
```

Going the other direction

Question: What is the probability of getting at least 6 correct?

```
1 - pbinom(5, size = n, prob = p)
```

```
[1] 0.05440223
```

R has a shortcut for this. Adding `lower.tail = FALSE` as an argument flips the direction.

NOTE: When `lower.tail = FALSE` is used, R calculates the probability of values greater than the number specified. It does not include the specified number.

```
pbinom(5, size = n, prob = p, lower.tail = FALSE)
```

```
[1] 0.05440223
```

This will calculate the probability getting 6 or more questions correct.

Alternatively, you can take the sum of probabilities over the specified interval.

```
sum(dbinom(6:12, size = n, prob = p))
```

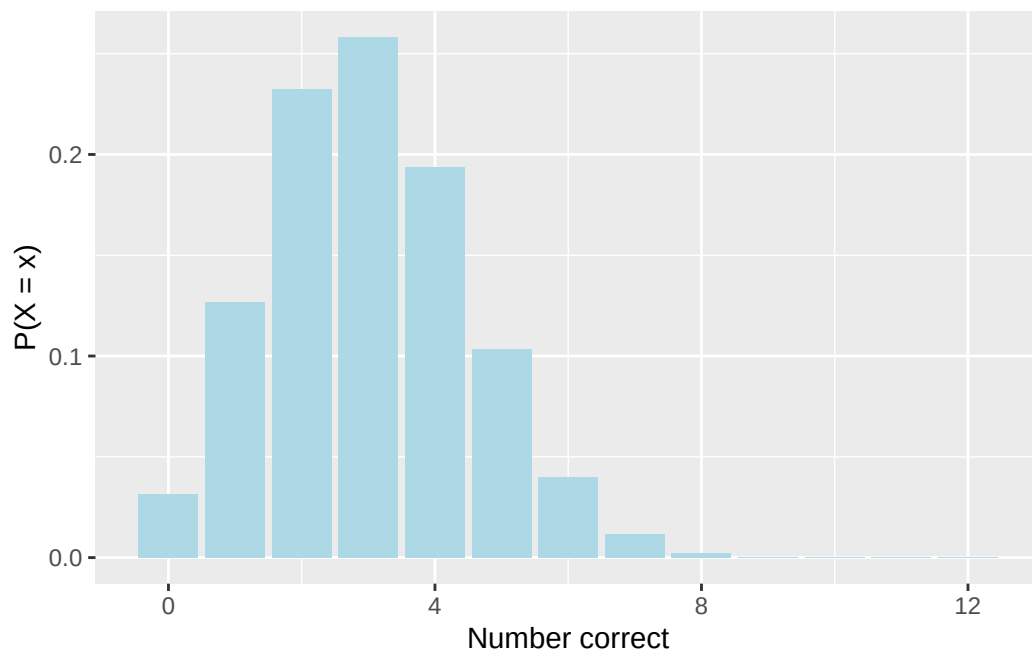
```
[1] 0.05440223
```

Seeing the distribution

Plotting dbinom for every outcome shows the shape

```
binom_df <- data.frame(x = 0:12)
binom_df$prob <- dbinom(binom_df$x, size = n, prob = p)

ggplot(binom_df, aes(x = x, y = prob)) +
  geom_col(fill = "lightblue") +
  labs(x = "Number correct", y = "P(X = x)")
```



qbinom and rbinom

qbinom runs backwards. Give it a probability and it returns the value

```
qbinom(0.90, size = n, prob = p)
```

```
[1] 5
```

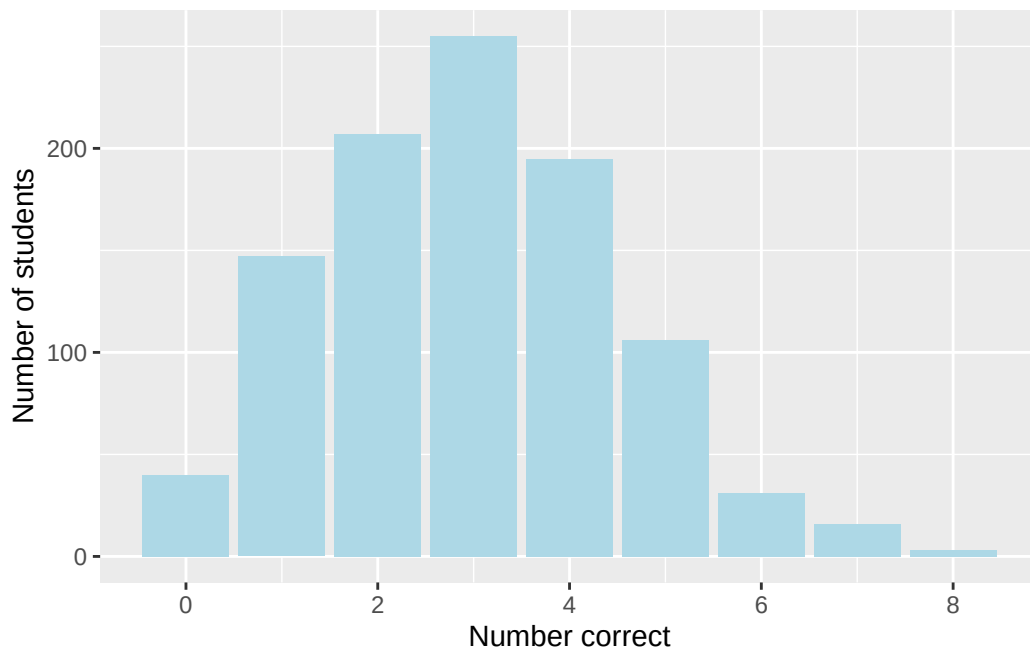
That says 90% of guessing students score 5 or fewer.

rbinom simulates. Here are 1000 imaginary students taking the quiz

```
set.seed(2389)

quizzes <- data.frame(correct = rbinom(1000, size = n, prob = p))

ggplot(quizzes, aes(x = correct)) +
  geom_bar(fill = "lightblue") +
  labs(x = "Number correct", y = "Number of students")
```



The Poisson distribution

The parameter, λ (lambda), the average number of events per interval.

A campus help line receives an average of 4 calls per hour.

$$X \sim \text{Pois}(4)$$

Note: $\lambda = 4$ is the expected number of calls in ONE hour. If you want the number of calls in a workday (8 hours), you must multiply by 8 to get that distribution: $\lambda = 32$.

```
lambda <- 4
```

dpois - exactly one value

Question: What is the probability of exactly 3 calls in an hour?

In symbols: $P(X = 3)$

```
dpois(3, lambda = lambda)
```

```
[1] 0.1953668
```

About a 19.5% chance of exactly 3 calls

ppois - a value or anything smaller

Question: What is the probability of 2 or fewer calls?

In symbols: $P(X \leq 2)$

```
ppois(2, lambda = lambda)
```

```
[1] 0.2381033
```

Question: What is the probability of more than 6 calls?

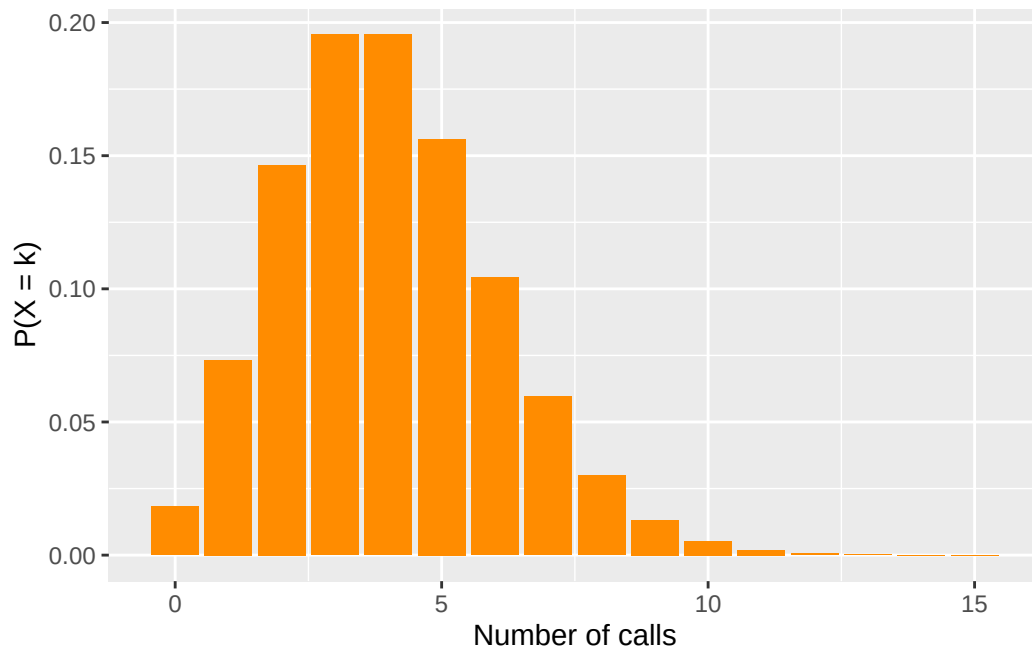
```
ppois(6, lambda = lambda, lower.tail = FALSE)
```

```
[1] 0.110674
```

Seeing the distribution

```
pois_df <- data.frame(k = 0:15)
pois_df$prob <- dpois(pois_df$k, lambda = lambda)

ggplot(pois_df, aes(x = k, y = prob)) +
  geom_col(fill = "darkorange") +
  labs(x = "Number of calls", y = "P(X = k)")
```



qpois and rpois

```
qpois(0.95, lambda = lambda)
```

```
[1] 8
```

```
calls <- rpois(1000, lambda = lambda)
```

The Normal distribution

The normal takes two parameters, the mean μ and the variance σ^2 . In R these are the arguments mean and sd.

Note: sd is the standard deviation, not the variance. If a problem tells you the variance is 25, you must take the square root and pass $\text{sd} = 5$

What changes for a continuous distribution

Because a measurement can land on any value, the probability of hitting one exact number is zero. $P(X = 70)$ is 0

Two consequences:

- Every normal question is about a range, never a single value.
- `dnorm` is not a probability. It's only the height of the curve, useful for drawing

Adult male heights are approximately normal with a mean of 70 inches and a standard deviation of 3 inches.

$$X \sim N(70, 9)$$

```
mu <- 70  
sigma <- 3
```

`pnorm` - the area to the left

Question: What proportion of men are shorter than 66 inches?

In symbols: $P(X < 66)$

```
pnorm(66, mean = mu, sd = sigma)
```

```
[1] 0.09121122
```

Answer: About 9% of men are shorter than 66 inches.

Question: What proportion are taller than 74 inches?

```
pnorm(74, mean = mu, sd = sigma, lower.tail = FALSE)
```

```
[1] 0.09121122
```

Question: What proportion are between 66 and 74 inches?

Take everything below 74 and subtract off everything below 66. What remains is everything between 74 and 66

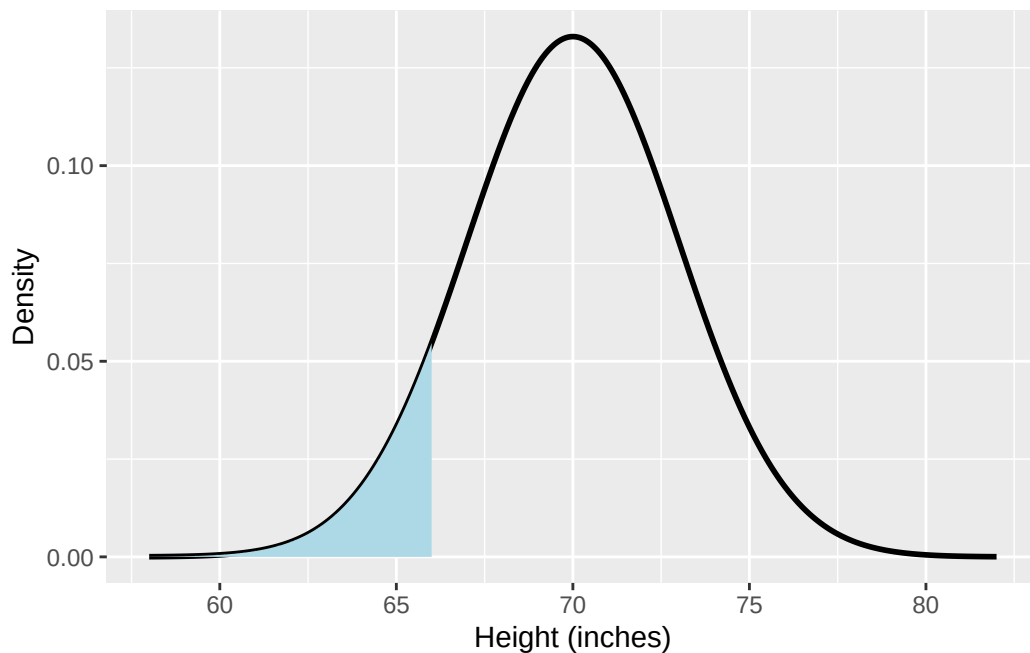
```
pnorm(74, mean = mu, sd = sigma) - pnorm(66, mean = mu, sd = sigma)
```

```
[1] 0.8175776
```

Seeing the distribution

```
curve_df <- data.frame(x = seq(58, 82, length.out = 1000))
curve_df$density <- dnorm(curve_df$x, mean = mu, sd = sigma)

ggplot(curve_df, aes(x = x, y = density)) +
  geom_line(linewidth=1) +
  geom_area(data = subset(curve_df, x <= 66), fill = "lightblue") +
  labs(x = "Height (inches)", y = "Density")
```



qnorm - going backwards from a probability

Question: How tall is a man at the 90th percentile?

```
qnorm(0.90, mean = mu, sd = sigma)
```

```
[1] 73.84465
```

Answer: A man at the 90th percentile is about 73.8 inches tall.

rnorm - simulation

```
heights <- rnorm(1000, mean = mu, sd = sigma)
mean(heights)
```

```
[1] 70.04687
```

```
sd(heights)
```

```
[1] 3.086093
```

The mean and sd of the simulated normal distribution are very close to the true parameters

Practice Problems

Answer each question in the code chunk provided. Where a problem asks you to state a distribution, write it in LaTeX using the notation from the lab.

Problem 1:

A seed company claims 85% of its seeds germinate. A gardener plants 20 seeds. Let X be the number that germinate.

- (a) State the distribution of X .

$$X \sim \text{Bin}(20, 0.85)$$

- (b) What is the probability that exactly 18 seeds germinate?

```
n <- 20
p <- 0.85

dbinom(18, n, p)
```

```
[1] 0.2293384
```

- (c) What is the probability that at least 18 germinate?

```
pbinom(17, n, p, lower.tail=FALSE)
```

```
[1] 0.4048963
```

- (d) What is the probability that 15 or fewer germinate?

```
pbinom(15, n, p)
```

```
[1] 0.1701532
```

(e) What is the probability that between 16 and 19 germinate?

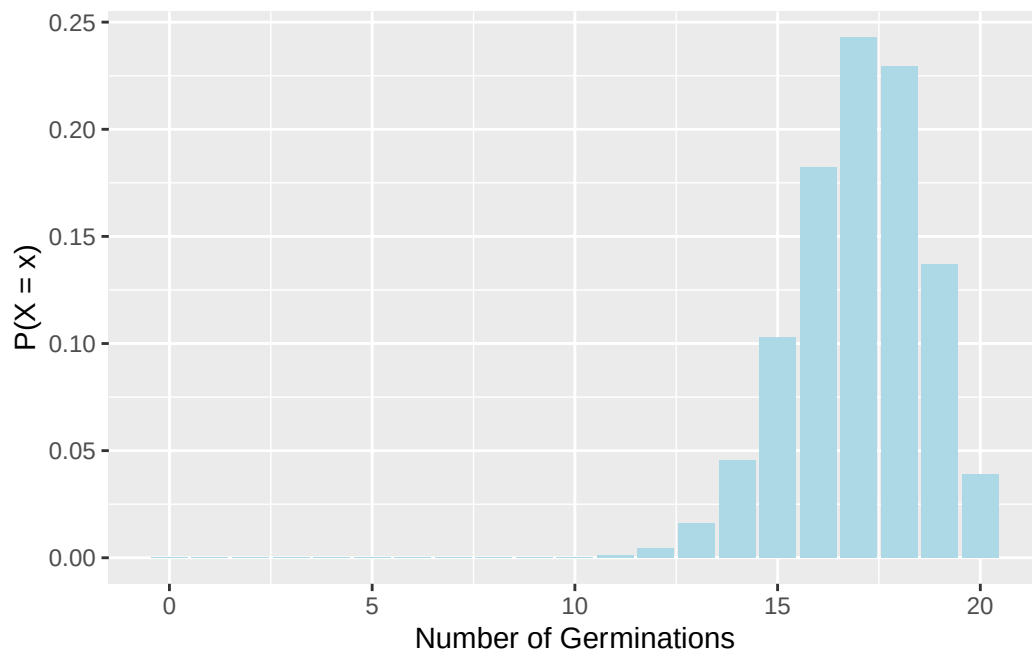
```
sum(dbinom(16:19, n, p))
```

```
[1] 0.7910873
```

(f) Use ggplot2 to make a bar plot of the distribution.

```
binom_df <- data.frame(x = 0:20)
binom_df$prob <- dbinom(binom_df$x, size = n, prob = p)

ggplot(binom_df, aes(x = x, y = prob)) +
  geom_col(fill = "lightblue") +
  labs(x = "Number of Germinations", y = "P(X = x)")
```



Problem 2:

A busy intersection averages 3.2 accidents per month.

- (a) State the distribution of the monthly accident count X

$$X \sim \text{Pois}(3.2)$$

- (b) What is the probability of exactly 5 accidents in a month?

```
lambda <- 3.2  
dpois(5, lambda)
```

```
[1] 0.1139794
```

- (c) What is the probability of no accidents in a month?

```
dpois(0, lambda)
```

```
[1] 0.0407622
```

- (d) What is the probability of at least 4 accidents in a month?

```
ppois(3, lambda, lower.tail=FALSE )
```

```
[1] 0.3974803
```

- (e) What is the probability of more than 40 accidents in a year? Remember to rescale λ

```
ppois(40, lambda*12, lower.tail=FALSE)
```

```
[1] 0.3584522
```

- (f) The city wants enough emergency response capacity to handle the monthly accident count in 99% of months. What number should they plan for? (Hint: you are finding a value, not a probability)

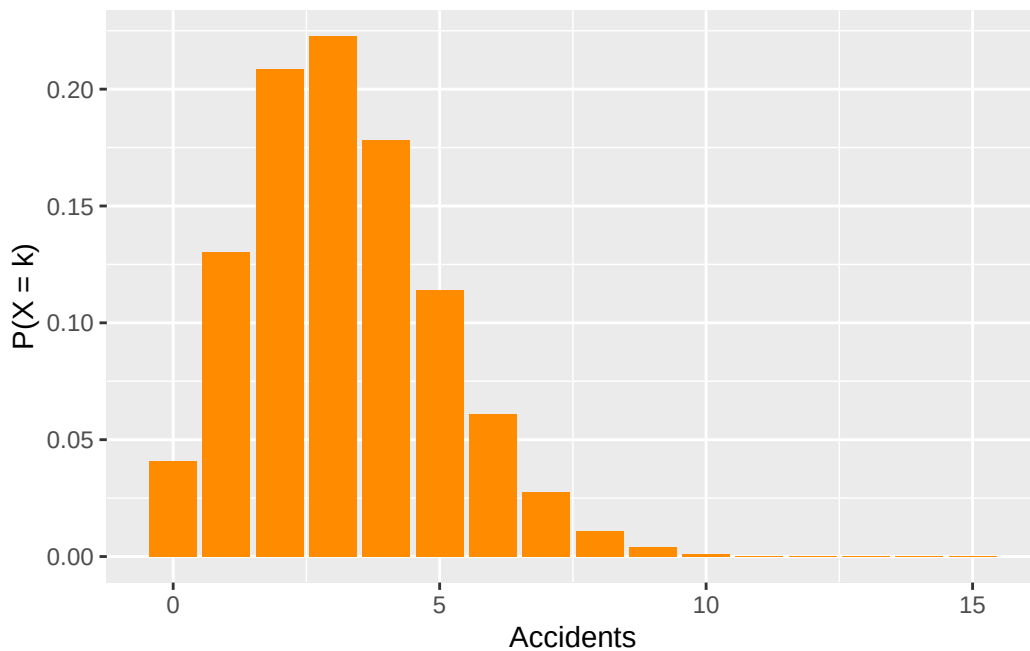
```
qpois(0.99,lambda)
```

```
[1] 8
```

(g) Use ggplot2 to make a plot of the distribution

```
pois_df <- data.frame(k = 0:15)
pois_df$prob <- dpois(pois_df$k, lambda = lambda)

ggplot(pois_df, aes(x = k, y = prob)) +
  geom_col(fill = "darkorange") +
  labs(x = "Accidents", y = "P(X = k)")
```



Problem 3:

A manufacturer's batteries last an average of 500 hours with a standard deviation of 45 hours, and battery life is approximately normally distributed.

(a) State the distribution of battery life X .

$$X \sim N(500, 45^2)$$

(b) What is the probability a battery lasts less than 425 hours?

```
mu <- 500
sigma <- 45

pnorm(425, mu, sigma)
```

```
[1] 0.04779035
```

(c) What is the probability a battery lasts more than 560 hours?

```
pnorm(560, mu, sigma, lower.tail=FALSE)
```

```
[1] 0.09121122
```

(d) What proportion of batteries last between 450 and 550 hours?

```
pnorm(550, mu, sigma) - pnorm(450, mu, sigma)
```

```
[1] 0.7334795
```

(e) The company will replace the shortest-lasting 2% of batteries under warranty. What lifetime should the warranty cover?

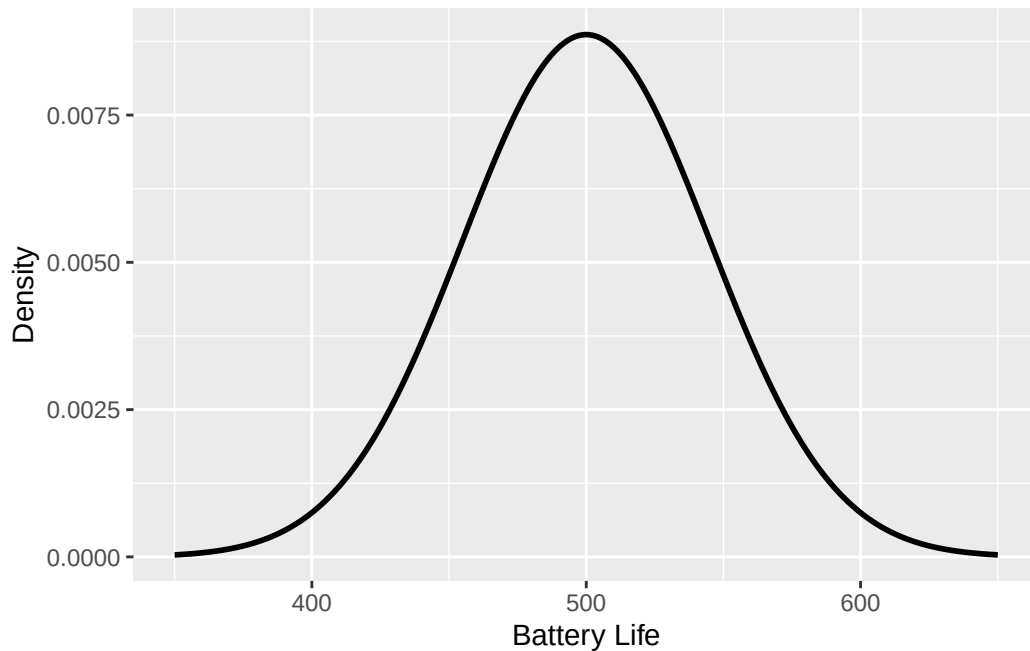
```
qnorm(0.02, mu, sigma)
```

```
[1] 407.5813
```

(f) Use ggplot2 to plot the distribution of battery life. (Hint: build a data frame of x values from 350 to 650, use dnorm for the curve height, and draw it with geom_line.)

```
curve_df <- data.frame(x = seq(350, 650, length.out = 1000))
curve_df$density <- dnorm(curve_df$x, mean = mu, sd = sigma)

ggplot(curve_df, aes(x = x, y = density)) +
  geom_line(linewidth=1) +
  labs(x = "Battery Life", y = "Density")
```



Problem 4: Choosing the right distribution

For each scenario, name the distribution and compute the requested probability.

- (a) A basketball player makes 78% of her free throws. She shoots 10 in a game. What is the probability she makes exactly 9?

Binomial

```
dbinom(9, 10, 0.78)
```

```
[1] 0.2351116
```

- (b) A website receives an average of 12 visits per minute. What is the probability it receives fewer than 8 visits in a given minute?

Poisson

```
ppois(7, 12)
```

```
[1] 0.0895045
```

- (c) Scores on a standardized exam are normally distributed with a mean of 1050 and a variance of 40000. What score is at the 75th percentile?

Normal

```
qnorm(0.75, 1050, 200)
```

```
[1] 1184.898
```

Final Step

Render your document. Click the Render button at the top of the editor, or press Ctrl + Shift + K.

If the render fails, read the error message. It names the chunk that caused the problem. The most common causes are a missing closing parenthesis, or an object was used before it was created.